

From Machine to Machine

An OCT-Trained Deep Learning Algorithm for Objective Quantification of Glaucomatous Damage in Fundus Photographs

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Purpose: Previous approaches using deep learning (DL) algorithms to classify glaucomatous damage on fundus photographs have been limited by the requirement for human labeling of a reference training set. We propose a new approach using quantitative spectral-domain (SD) OCT data to train a DL algorithm to quantify glaucomatous structural damage on optic disc photographs.

Design: Cross-sectional study.

Participants: A total of 32 820 pairs of optic disc photographs and SD OCT retinal nerve fiber layer (RNFL) scans from 2312 eyes of 1198 participants.

Methods: The sample was divided randomly into validation plus training (80%) and test (20%) sets, with randomization performed at the patient level. A DL convolutional neural network was trained to assess optic disc photographs and predict SD OCT average RNFL thickness.

Main Outcome Measures: The DL algorithm performance was evaluated in the test sample by evaluating correlation and agreement between the predictions and actual SD OCT measurements. We also assessed the ability to discriminate eyes with glaucomatous visual field loss from healthy eyes with area under the receiver operating characteristic (ROC) curves.

Results: The mean prediction of average RNFL thickness from all 6292 optic disc photographs in the test set was 83.3 ± 14.5 μm , whereas the mean average RNFL thickness from all corresponding SD OCT scans was 82.5 ± 16.8 μm ($P = 0.164$). There was a very strong correlation between predicted and observed RNFL thickness values (Pearson $r = 0.832$; $R^2 = 69.3\%$; $P < 0.001$), with mean absolute error of the predictions of 7.39 μm . The area under the ROC curves for discriminating glaucomatous from healthy eyes with the DL predictions and actual SD OCT average RNFL thickness measurements were 0.944 (95% confidence interval [CI], 0.912 – 0.966) and 0.940 (95% CI, 0.902 – 0.966), respectively ($P = 0.724$).

Conclusions: We introduced a novel DL approach to assess fundus photographs and provide quantitative information about the amount of neural damage that can be used to diagnose and stage glaucoma. In addition, training neural networks to predict SD OCT data objectively represents a new approach that overcomes limitations of human labeling and could be useful in other areas of ophthalmology. *Ophthalmology* 2019;126:513–521 © 2018 by the American Academy of Ophthalmology



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Glaucoma is a progressive optic neuropathy that results in characteristic changes to the optic disc and retinal nerve fiber layer (RNFL).¹ Although damage from glaucoma is irreversible, early treatment usually can prevent or slow down progression to functional damage and visual impairment.² Assessment of structural damage is essential for early detection of glaucoma. Fundus photographs are a low-cost and easy-to-perform method to document and identify optic disc features characteristic of glaucoma. However, it is well established that subjective evaluation of optic disc photographs suffers from low reproducibility, even when performed by expert graders.^{3–5} In addition, graders frequently underestimate

or overestimate glaucoma likelihood when evaluating disc photographs.⁶ Recent progress in artificial intelligence and machine learning has led to the development of algorithms capable of objective assessment of fundus photographs for identification of signs of ocular diseases.^{7–11} Li et al⁷ evaluated the ability of a deep learning (DL) neural network algorithm to identify signs of glaucomatous neuropathy on color fundus photographs. The authors reported excellent sensitivity and specificity to diagnose “referable” glaucomatous optic neuropathy, which was defined based on subjective grading of the photographs by a group of trained ophthalmologists.

A fundamental step in the development of any machine learning algorithm is the training process by which the algorithm “learns” to make classifications and predictions correctly. Essentially, the algorithm cannot perform better than the reference standard used to train it, and its best hope is to replicate perfectly the classifications or predictions that would have been made by the reference standard. Although the work by Li et al⁷ provides important insights into how machine learning could be used to assess glaucomatous damage on fundus photographs, their algorithm suffers from the limitation that subjective gradings were used as the reference standard to train the DL network.

In recent decades, spectral-domain (SD) OCT has become the de facto standard in objective quantification of structural damage in glaucoma.¹² Measurements of the RNFL thickness with SD OCT demonstrate high reproducibility and have been shown to diagnose the disease accurately, to detect its progression, and to measure rates of change.^{12–15} As such, it is conceivable that an accurate machine learning classifier could be obtained if it is trained to evaluate fundus photographs using SD OCT measurements as the reference standard, rather than subjective assessment by graders. Automated SD OCT measurements are easier to obtain and thus may make training of classifiers on large data sets more feasible. Such an algorithm also could be trained to obtain quantitative rather than just qualitative assessments of the amount of neural damage from disc photographs.

In the present work, we examined a novel DL algorithm trained to assess optic disc photographs from results of SD OCT and investigated its ability to provide objective quantification of glaucomatous neural loss.

Methods

The data set for this study was collected from the Duke Glaucoma Repository, a database of electronic medical and research records at the Vision, Imaging and Performance Laboratory of the Duke Eye Center. The institutional review board from Duke University approved this study, and a waiver of informed consent was provided because of the retrospective nature of this work. All methods adhered to the tenets of the Declaration of Helsinki for research involving human participants, and the study was conducted in accordance with regulations of the Health Insurance Portability and Accountability Act.

The database contained information on comprehensive ophthalmologic examinations during follow-up, diagnoses, medical history, visual acuity, slit-lamp biomicroscopy, intraocular pressure measurements, and results of gonioscopy and dilated slit-lamp funduscopy examinations. In addition, the repository contained stereoscopic optic disc photographs (Nidek 3DX; Nidek, Gamagori, Japan) and standard automated perimetry (SAP; Humphrey Field Analyzer II; Carl Zeiss Meditec, Inc., Dublin, CA) and Spectralis SD OCT (software version 5.4.7.0; Heidelberg Engineering, GmbH, Dossenheim, Germany) images and data. Standard automated perimetry was acquired with the 24-2 Swedish interactive threshold algorithm (Carl Zeiss Meditec, Inc.). Only participants with open angles on gonioscopy were included. Visual fields were excluded if they had more than 33% fixation losses or more than 15% false-positive errors. Patients were excluded if they had a history of other ocular or systemic diseases that could affect the optic nerve or the visual field.

Diagnosis of glaucoma was defined based on the presence of glaucomatous repeatable visual field loss in SAP (pattern standard deviation <5% or glaucoma hemifield test results outside normal limits) and signs of glaucomatous optic neuropathy as based on records of slit-lamp fundus examination. Glaucoma suspects were those with a history of elevated intraocular pressure, suspicious appearance of the optic disc on slit-lamp fundus examination, or other risk factors for the disease. Healthy participants were required to have a normal optic disc appearance on slit-lamp fundus examination in both eyes as well as no history of elevated intraocular pressure and normal SAP results.

Images were acquired with the Spectralis SD OCT to assess the RNFL. The device uses a dual-beam SD OCT and a confocal scanning laser ophthalmoscope that uses a super luminescent diode light with a center wavelength of 870 nm and an infrared scan to provide simultaneous images of ocular microstructures. The Spectralis RNFL circle scan was used for this study. A total of 1535 A-scan points were acquired from a 3.45-mm circle centered on the optic disc. Axial length and corneal curvature measurements were entered into the instrument software to ensure accurate scaling of all measurements, and the device’s eye-tracking capability was used during image acquisition to adjust for eye movements and to ensure that the same location of the retina was scanned over time. Images were reviewed manually to ensure quality, scan centration, and no coexistent retinal pathologic abnormalities. Images that had been inverted or clipped or with signal strength less than 15 dB were excluded. The average circumferential RNFL thickness corresponds to the 360° measure automatically calculated by the SD OCT software.

For each eye of each participant, we considered all the available optic disc photographs that had been acquired over time and matched them to the closest Spectralis SD OCT RNFL scan acquired within 6 months from the photograph date. Because participants were followed up over time, multiple pairs of SD OCT and disc photographs were available for each participant. This was important to increase the heterogeneity of the data set for DL training.

Development of the Deep Learning Algorithm

A DL algorithm was trained to predict SD OCT average RNFL thickness from assessment of optic disc photographs. The target value, that is, the variable we wanted to predict from analysis of optic disc photographs, was the SD OCT average RNFL thickness. Therefore, for training the neural network, a pair of train-target consisted of the optic disc photograph and the SD OCT average RNFL thickness value. The sample of pairs of photographs and OCT scans was split into a training plus validation set (80%) and test sample (20%). Importantly, to prevent leakage and biased estimates of test performance, the random sampling process was at the patient level, so no data of any patient were present in both the training and the test samples.

The optic disc stereophotographs initially were preprocessed to derive data for the DL algorithm. Each stereoscopic photograph was split, creating a pair of photographs from the stereo views. The images then were downsampled to 256×256 pixels and pixel values were scaled to range from 0 to 1. Data augmentation was performed to increase heterogeneity of the photographs, reducing the possibility of overfitting and allowing the algorithm to learn the most relevant features. Data augmentation included the following: random lighting, consisting of subtle changes in image brightness and contrast of up to 5%; random rotation, consisting of rotations of up to 10° in the image; and random flips, consisting of flipping the image vertically or horizontally.

We used a residual deep neural Network (ResNet34) architecture. The ResNet is a revolutionary deep residual network that has

allowed relatively rapid training of very deep convolutional neural networks in a way that had not been possible previously.¹⁶ In brief, these networks use identity shortcut connections that skip 1 or more layers and greatly decrease the vanishing gradient problem when training deep networks. In the present work, a ResNet that previously had been trained on the ImageNet data set¹⁷ was used. However, because the recognition task of the present work differs largely from that of ImageNet, further training was performed by initially unfreezing the last 2 layers. Subsequently, all layers were unfrozen, and training was performed using differential learning rates, where different learning rates are used for different parts of the network, with a lower rate for the earlier layers and gradually increasing it for the later layers. The network was trained with minibatches of size 64 and Adam optimizer.^{18,19} The best learning rate was found using the cyclical learning method with stochastic gradient descent with restarts.²⁰

As a variant of the training process described above, we also trained the DL network to classify optic disc photographs as normal versus abnormal according to the SD OCT average RNFL thickness categorical classification as provided by the instrument's normative database. This was done to allow investigation of how DL assessment of photographs would perform in classifying and categorizing the presence of damage. The SD OCT instrument classifies the average RNFL thickness in 1 of 3 possible categories by comparing the measurement with values from the instrument's normative database. These 3 categories are normal, borderline, and abnormal. In our analysis, we collapsed the borderline and normal categories into the normal one so that we had a binary target variable (normal vs. abnormal) that retained high specificity. The DL model then calculated the probability of abnormality based on assessment of optic disc photographs. We built heatmaps corresponding to the gradient-weighted class activation maps over the input images.^{21,22} These heatmaps indicate how important each location of the image is with respect to the class under consideration. This technique allows one to visualize the parts of the image that are most important in the deep neural network classification.

Statistical Analyses

The performance of the DL algorithm in quantifying glaucomatous damage in optic disc photographs was evaluated in the test sample by comparing the predictions with the actual SD OCT average RNFL thickness. Generalized estimating equations (GEEs) were used to account for the fact that multiple measures were obtained per patient.²³ We calculated the mean absolute error of the predictions as well as the Pearson correlation coefficient and agreement by the Bland-Altman plot and 95% confidence limits of agreement. We also investigated the correspondence between classifications performed by the DL system and those given by the SD OCT normative database.

We also investigated the relationship between predicted and observed values of RNFL thickness and SAP mean deviation with locally weighted scatterplot smoothing.²⁴ The closest visual field to the SD OCT was used to assess the structure–function relationship. Receiver operating characteristic (ROC) curves were used to assess and compare the ability of the DL algorithm on photographs versus SD OCT average RNFL thickness in discriminating eyes with glaucoma from healthy eyes, as defined above for inclusion in the study. The ROC curve provides the tradeoff between the sensitivity and 1 – specificity. The area under the ROC curve was used to summarize the diagnostic accuracy of each parameter. An area under the ROC curve of 1.0 represents perfect discrimination, whereas an area of 0.5 represents chance discrimination. Sensitivity at fixed specificities of 80% and 95% also were reported. To account for using multiple images of both eyes of the same

participant in the analyses, a bootstrap resampling procedure was used to derive confidence intervals and *P* values, where the cluster of data for the participant was considered as the unit of resampling to adjust standard errors. This procedure has been used previously to adjust for the presence of multiple correlated measurements from the same unit.²⁵

Results

The data set included 32 820 pairs of optic disc photographs and SD OCT scans from 2312 eyes of 1198 participants. The test sample consisted of 6292 pairs of disc photographs and SD OCT scans from 463 eyes of 240 participants. Table 1 shows demographic and clinical characteristics of the participants and eyes in the training and test samples.

Figure 1 shows the relationship between DL predictions of average RNFL thickness from optic disc photographs and the actual SD OCT measurements in the test sample. The mean prediction of average RNFL thickness from all 6292 optic disc photographs was 83.3 ± 14.5 μm , whereas the mean average RNFL thickness from all the 6292 corresponding SD OCT scans was 82.5 ± 16.8 μm ($P = 0.164$, GEE). There was a very strong correlation between the predicted and the observed RNFL thickness values (Pearson $r = 0.832$; $R^2 = 69.3\%$; $P < 0.001$), with mean absolute error of 7.39 μm . Figure S2 (available at www.aaojournal.org) shows the Bland-Altman plot assessing the agreement between predictions and observations. The 95% confidence limits of agreement ranged from -18.5 μm to 17.5 μm , with no statistically significant evidence of proportional bias ($P = 0.074$). Figure 3 shows violin plots illustrating the distribution of predicted and observed RNFL thickness values in normal, suspect, and glaucomatous eyes in the test sample. Average predictions were 96.1 ± 7.8 , 87.5 ± 9.9 , and 71.0 ± 14.4 μm in normal, suspect, and glaucomatous eyes, respectively. There was a statistically significant difference between average predictions for all pairwise comparisons among the 3 groups ($P < 0.001$, GEE). Corresponding numbers for the mean SD OCT average RNFL thickness in the 3 groups were 97.6 ± 9.3 , 87.1 ± 12.5 , and 68.8 ± 16.0 μm ($P < 0.001$ for all pairwise comparisons, GEE).

Figure S4 (available at www.aaojournal.org) illustrates the relationship between SAP mean deviation versus observed and predicted RNFL thickness values. The area under the ROC curve for discriminating glaucomatous from normal eyes with the DL optic disc photograph predictions was 0.944 (95% confidence interval [CI], 0.912–0.966), whereas the area under the ROC curve for actual SD OCT average RNFL thickness was 0.940 (95% CI, 0.902–0.966). There was no statistically significant difference between the areas under the ROC curves ($P = 0.724$). For specificity at 95%, the predicted measurements had sensitivity of 76% (95% CI, 64%–84%), whereas actual SD OCT measurements had sensitivity of 73% (95% CI, 59%–85%). For specificity at 80%, the predicted measurements had sensitivity of 90% (95% CI, 82%–95%), whereas actual SD OCT measurements had sensitivity of 90% (95% CI, 83%–95%).

From the 6292 OCT scans in the test set, 1908 (30.3%) were classified as abnormal and 4384 (69.7%) as normal, according to the instrument's normative database. The DL network trained on photographs achieved an overall accuracy of 83.7% to replicate such classification. Figure 5 shows examples of optic disc photographs and corresponding class activation maps (heatmaps) of the DL network. The heatmaps show that the activations were found most strongly in the area of the optic nerve and adjacent RNFL on the photographs, indicating that these areas were the most important for the network classifications. Figures 6 and 7

Table 1. Demographic and Clinical Characteristics of the Eyes and Participants Included in the Training and Test Samples

	Healthy	Suspect	Glaucoma
Training sample*			
No. of eyes	476	674	699
No. of images	3982	13 410	9136
Age (yrs)	57.8±13.9	65.4±11.1	69.7±11.2
Female gender (%)	64.7	60.5	53.1
Race (%)			
White	56.7	61.8	60.2
Black	43.3	38.2	39.8
SAP MD (dB)	0.05±1.10	−0.62±1.91	−7.37±6.95
SAP PSD (dB)	1.60±0.40	1.94±1.10	6.40±3.94
SD OCT average RNFL thickness (μm)	96.8±10.9	89.1±12.8	68.3±14.8
Test sample†			
No. of eyes	128	164	171
No. of images	877	3345	2070
Age (yrs)	56.5±15.9	65.5±11.3	68.1±12.8
Female gender (%)	59.1	64.5	45.2
Race (%)			
White	51.8	65.8	58.0
Black	48.2	34.2	42.0
SAP MD (dB)	−0.06±1.10	−0.62±2.36	−7.65±6.9
SAP PSD (dB)	1.61±0.35	2.00±1.19	6.63±3.99
SD OCT average RNFL thickness (μm)	97.6±9.3	87.1±12.5	68.8±16.0

MD = mean deviation; PSD = pattern standard deviation; RNFL = retinal nerve fiber layer; SAP = standard automated perimetry; SD = spectral-domain.

Data are mean ± standard deviation unless otherwise indicated.

*Comprising 26 528 pairs of disc photographs and SD OCT scans from 1849 eyes of 958 participants.

†Comprising 6292 pairs of disc photographs and SD OCT scans from 463 eyes of 240 participants.

show several random examples of optic disc photographs from the test sample where the DL algorithm correctly and incorrectly predicted the classification given by the SD OCT average RNFL thickness, respectively.

Discussion

In the present study, we developed and validated a novel DL algorithm to assess optic disc photographs for the presence of glaucomatous damage. In contrast to previous works in this area, our algorithm was capable of outputting continuous predictions of estimated RNFL thickness, therefore allowing for quantitative assessment of the amount of neural damage on disc photographs. This was achieved by training the network with RNFL thickness measurements extracted from SD OCT. To the best of our knowledge, such an approach has not been described previously in the literature.

Previous investigators have described machine learning approaches to assess optic disc photographs for glaucomatous damage.^{7,9} In these investigations, human graders were asked to label the photographs for the presence of glaucomatous damage, and this labeling was used as the reference standard to train the classifier. Such an approach presents several limitations. Gratings of optic disc

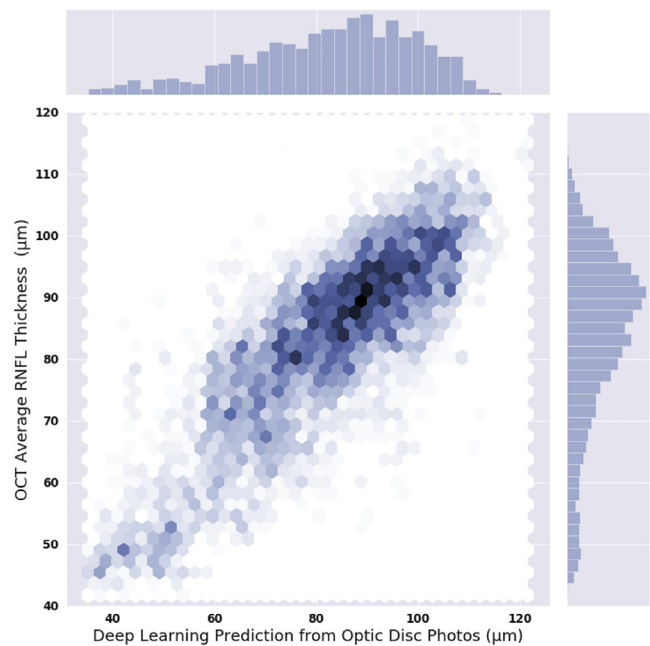


Figure 1. Scatterplot and histograms illustrating the relationship between predictions obtained by the deep learning algorithm evaluating optic disc photographs and actual average retinal nerve fiber layer (RNFL) thickness measurements from spectral-domain OCT. Data are from the independent test set.

photographs by human graders are subjective and known to have relatively poor reproducibility.^{3–5} Furthermore, misclassifications are very likely to occur. For example, graders frequently tend to misclassify eyes with physiologic large cups as having glaucoma and they often miss signs of glaucomatous damage in eyes with small optic discs. If a machine learning classifier is trained using human labeling of optic disc photographs, it will essentially replicate those errors and is likely to have poor performance as a screening tool, even if it shows high accuracy when compared with the human gradings. Our study proposes a novel approach in training the classifier by using RNFL thickness measurements extracted from SD OCT. This presents some obvious advantages. First, it provides an objective and reproducible metric to serve as a target. Average RNFL thickness measurements have been shown to detect glaucomatous damage accurately and, for example, can discriminate eyes with large physiologic cups from those with actual glaucomatous damage.^{26–28} In addition, training a network with objective SD OCT data obviates the need for the time-consuming task of subjective labeling by human graders.

The predictions obtained from DL analysis of optic disc photographs showed very strong correlation with the actual RNFL thickness measurements in the independent test sample. Furthermore, the mean absolute error of the predictions was only approximately 7 μm. Interestingly, a previous study reported an R^2 of 70% for the correlation between 2 different SD OCT devices, and another study reported Bland-Altman limits of agreement that are not far from those reported in our investigation.^{29,30} As a result, it was not surprising that the predictions performed well to

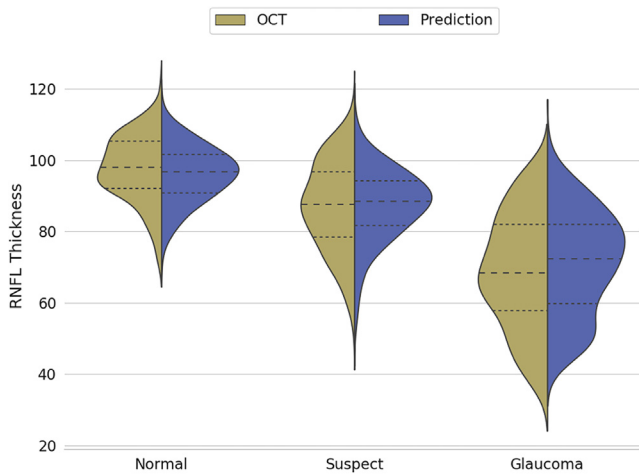


Figure 3. Violin plots illustrating the distribution of deep learning predictions and OCT average retinal nerve fiber layer (RNFL) thickness in normal, suspect, and glaucomatous eyes.

discriminate eyes with glaucomatous visual field loss from healthy eyes. In fact, the area under the ROC curve for the predictions was almost identical to that for actual SD OCT RNFL thickness values. This provides important validation of our DL model. As described before, previous models essentially have attempted to replicate human grading of photographs but have provided no evidence that the algorithm classifications or predictions actually corresponded to clinically relevant outcomes. By showing a correspondence to visual field loss in a similar degree to SD OCT, our work provides essential validation of the quantitative DL approach to assess disc photographs.

Although SD OCT has become the reference standard for quantification of structural damage in glaucoma, assessment of optic disc photographs may present several advantages. Spectral-domain OCT images generally remain expensive and the machines are nonportable, which can create conditions making it difficult to implement OCT in screening settings, especially in underserved populations. In contrast, photographs may provide a quick and inexpensive method for documenting the optic disc appearance. Recent work has demonstrated the feasibility of acquiring fundus photographs with portable devices and cell phones.^{31–33} Although using SD OCT may remain relatively unfeasible in most screening settings, our approach demonstrates that a DL algorithm can replicate closely general SD OCT average RNFL thickness measurements from optic disc photographs and potentially could be implemented in low-cost screening settings using fundus photographs. Further investigations should evaluate the feasibility and accuracy of such approaches.

Another important advantage of the quantitative approach for assessing optic disc photographs presented in this work is the potential for assessing changes over time in settings where SD OCT is not available. A qualitative yes-or-no assessment of disc photographs as performed previously generally does not allow assessment of changes over time, notably in those already classified as glaucomatous at baseline. By providing a continuous output, our approach potentially could be used to extract progression information from optic disc photographs that could be used for monitoring glaucomatous damage. However, validation of such an approach will require longitudinal investigations.

The activation heatmaps showed that the locations in the optic disc photographs that were most important for the DL algorithm corresponded very closely to the optic disc and adjacent RNFL, as seen in Figure 5. Retinal blood vessels or

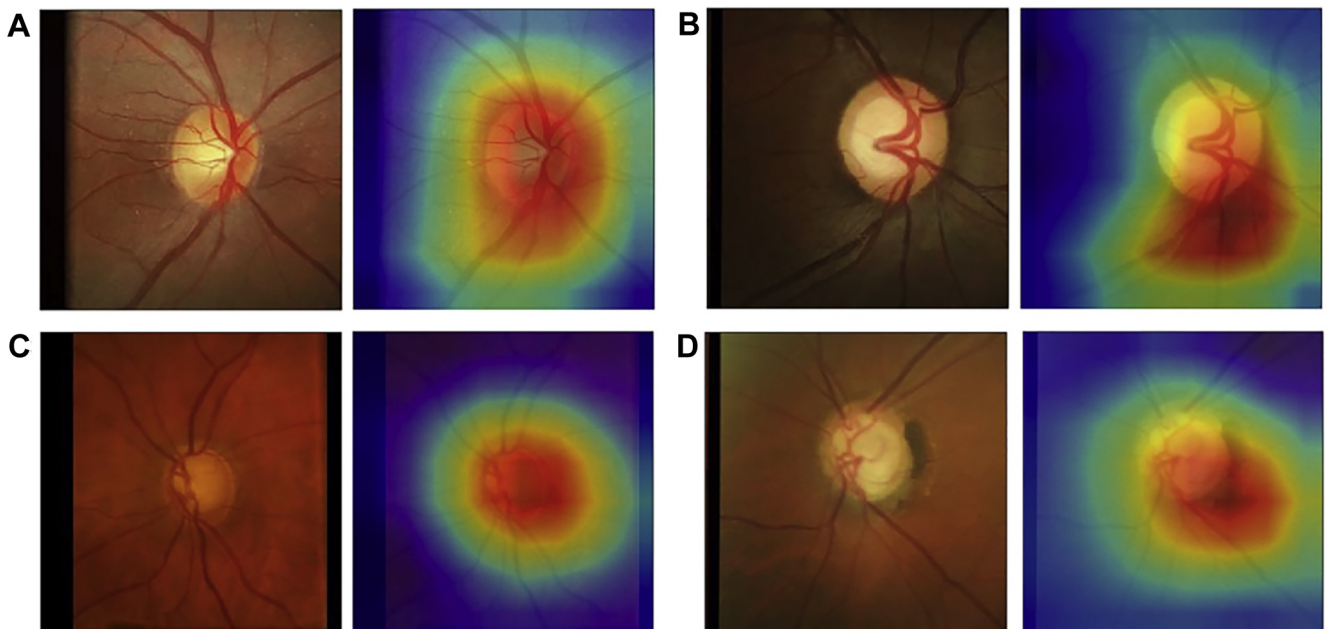


Figure 5. Class activation maps (heatmaps) showing the regions of the photograph that had greatest weight in the deep learning algorithm classification. A, Healthy eye. B, Glaucoma suspect. C, D, Glaucomatous eyes.



Figure 6. Random examples of optic disc photographs that were classified correctly according to the reference classification of the Spectralis spectral-domain OCT normative database for average retinal nerve fiber layer (RNFL) thickness. Above each photograph is shown the OCT average RNFL thickness measurement, the deep learning (DL) prediction of average RNFL thickness from the optic disc photograph, and the probability of abnormality estimated by the DL algorithm.



Figure 7. Random examples of optic disc photographs that were classified incorrectly according to the reference classification of the Spectralis spectral-domain OCT normative database for average retinal nerve fiber layer (RNFL) thickness. Above each photograph is shown the OCT average RNFL thickness measurement, the deep learning (DL) prediction of average RNFL thickness from the optic disc photograph, and the probability of abnormality estimated by the DL algorithm.

areas farther from the optic disc had much smaller activations. This provides further confirmation that the algorithm is indeed identifying the area of the photograph that is important for diagnosing glaucoma. The approach presented here may allow future investigations of features of optic discs that present the greatest challenges for recognition of signs of the disease, increasing awareness for their significance and opening opportunities for better training of clinicians on how to recognize them in clinical practice.

A large proportion of our patient population was black. Population-based studies have demonstrated that black persons have an increased prevalence of primary open-angle glaucoma compared with white persons.^{34–36} Moreover, they may progress at a faster rate and experience a greater degree of functional impairment resulting from glaucoma at a younger age.^{36–38} Therefore, improved screening and testing methods are needed to target this potentially high-risk group. Our algorithm may be particularly useful in this regard and could find use in teleophthalmology programs targeting individuals at greatest risk for visual impairment. Further studies should investigate this approach and should incorporate images acquired from patients of other races and ethnicities to improve the generalizability of the algorithm.

Our study has limitations. It should be noted that although the DL algorithm presented in this study performed well in identifying glaucomatous damage, approximately 30% of the variance of SD OCT measurements remained unexplained. Many factors can explain this, such as variability of optic disc appearances and SD OCT measurements, as well as differences in photograph quality. In fact, it would be surprising to find higher correlations between photograph predictions and SD OCT, considering that SD OCT obtains precise measurements of tissue thickness at a micrometer scale. We also have not performed a qualitative assessment of the optic disc photographs for quality. It is possible that removing poor-quality photographs would result in even better performance of the algorithm. In fact, Figure 7 shows some random cases that were misclassified by the DL algorithm, and it is possible to see that some of them had relatively low-quality photographs. However, retention of disc photographs of lower quality could improve the generalizability of our model in its application to clinical and teleophthalmology settings. Furthermore, the algorithm was trained to replicate average thickness measurements from SD OCT rather than segmental SD OCT loss. It is likely that more sophisticated approaches could be created by having different SD OCT measurements as target values, including sectoral measurements from other areas of the optic disc or macula. In fact, in one of the examples shown in Figure 7 (second row, first photograph from the left), the DL algorithm classification (abnormal, with $P = 0.72$) disagreed with the SD OCT average RNFL classification (normal), but subjective analysis of the disc photograph actually shows inferior localized rim thinning. Further training of the DL network with bigger data sets is likely to improve its performance even further.

In conclusion, we introduced a novel DL approach to assess optic disc photographs and provide quantitative information about the amount of neural damage. By analyzing disc photographs, the DL algorithm was trained to

replicate closely measurements obtained from average SD OCT RNFL thickness. The approach presented in this work potentially could be used to diagnose and stage glaucomatous damage from optic disc photographs. In addition, it is possible that the innovative approach first proposed in this study of using OCT to train DL models may find use in other areas of ophthalmology.

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Footnotes and Financial Disclosures

Originally received: October 7, 2018.

Final revision: November 13, 2018.

Accepted: December 11, 2018.

Available online: December 20, 2018. Manuscript no. 2018-2269.

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Financial Disclosure(s):

The author(s) have made the following disclosure(s): F.A.M.: Consultant — Alcon Laboratories, Allergan, Carl Zeiss Meditec, Sensimed, Topcon, Reichert; Financial support — Alcon Laboratories, Allergan, Bausch & Lomb, Carl Zeiss Meditec, Heidelberg Engineering, Merck, nGoggle, Inc. Supported in part by the National Eye Institute, National Institutes of Health, Bethesda, Maryland (grant nos.: EY027651 [F.A.M.], EY025056 [F.A.M.], and EY021818 [F.A.M.]); the Conselho Nacional de Desenvolvimento Científico e Tecnológico (A.A.J.); and the Heed Foundation (A.C.T.). The funding organizations had no role in the design or conduct of this research.

HUMAN SUBJECTS: Human subjects were included in this study. The data set for this study was collected from the Duke Glaucoma Repository, a database of electronic medical and research records at the Vision, Imaging and Performance (VIP) Laboratory of the Duke Eye Center. The institutional review board at Duke University approved this study, and a waiver of

informed consent was provided due to the retrospective nature of this work. All methods adhered to the tenets of the Declaration of Helsinki for research involving human subjects and the study was conducted in accordance with regulations of the Health Insurance Portability and Accountability Act.

No animal subjects were included in this study.

Author Contributions:

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Data collection: Medeiros, Jammal, Thompson

Obtained funding: Medeiros

Overall responsibility: Medeiros, Jammal, Thompson

Abbreviations and Acronyms:

CI = confidence interval; **DL** = deep learning; **GEE** = generalized estimating equation; **ROC** = receiver operating characteristic; **RNFL** = retinal nerve fiber layer; **SAP** = standard automated perimetry; **SD** = spectral-domain.

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